

Seongsu Kim

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Research Interests

My research goal is to build real-world robotic agents for general dexterous manipulation. I am interested in integrating robot learning algorithms with real-world robotic systems to develop agents with physical intelligence for diverse, complex, and long-horizon tasks.

- Imitation learning via generative models
- World models
- Offline RL / Offline Goal-conditioned RL
- Sample-efficient RL fine-tuning

Education

Hanyang University

Mar 2023 – Feb 2025

M.S. in Electrical Engineering (Advisor: [Jun Moon](#))

Thesis: *"Model-based learning to achieve goals using offline data"*

GPA: 4.33 / 4.5

Dongyang Mirae University

Mar 2017 – Feb 2023

B.S. in Automation Engineering

Leave of absence for military service: Aug. 2017 - May. 2019

GPA: 4.29 / 4.5

Experience

Researcher

Jul 2025 - Present

[Korea Electronics Technology Institute \(KETI\)](#), Intelligent Robotics Research Center

- Developed visuo-tactile imitation learning policies for precise insertion tasks using Diffusion Policy and tactile sensing.
- Worked on bimanual dexterous manipulation with multifingered robotic hands for coordinated manipulation tasks.
- Investigated humanoid whole-body manipulation with egocentric vision for robot manipulation in complex environments.

Research Intern

Dec 2022 - Feb 2023

Control & Optimization Lab, Hanyang Univ.

Advisor: Prof. [Jun Moon](#)

- Constructed a framework for operating the Husky-Franka robot using Isaac-Sim and ROS

Publications

(*: equal contribution)

- [P1] Geunha Kim*, **Seongsu Kim***, Junbeom Lee, and Jun Moon. "Enhancing Vision-Language-Action model with Reward-free Goal-conditioned Reinforcement Learning for Long-Horizon Control" *Preprint*, 2026.

- [J1] Younjae Go*, Yuna Oh*, **Seongsu Kim**, and Jun Moon. “GASP: Generative Adversarial State Policy and Multi-modal Sensing Framework for Dexterous Manipulation” *International Journal of Control, Automation, and Systems*, 24, 1739–1751, 2026.
- [C5] Mingi Jung, **Seongsu Kim**, Seungwon Lee, and Jongbum Park. “A Standardized Test Method and Metric Benchmark for Imitation-Learning-Based Robotic Assembly” *Automation World 2026, International Conference On Consumer Electronics (ICCE) Asia*, 2026.
- [C4] **Seongsu Kim**, Jusung Kim, Mingi Jung, Seungwon Lee, and Jongbum Park. “Tactile-Guided Residual Reinforcement Learning for Precise Connector Insertion” *International Conference On Consumer Electronics (ICCE) Asia*, 2026.
- [C3] **Seongsu Kim**, Seungwon Lee, and Jongbum Park. “A Comparative Study of SDE-based Diffusion and ODE-based Flow Matching Policies for Precise Inserting in Imitation learning” *The 20th Korea Robotics Society Annual Conference (KRoC)*, 2025.
- [C2] **Seongsu Kim**, Geunha Kim, Dongbeom Kim, and Jun Moon. “Classify Objects with VLA Grasp Network” *International Conference on Control, Automation, and Systems (ICCAS)*, 2025.
- [C1] **Seongsu Kim** and Jun Moon. “Offline Goal-conditioned Model-based Reinforcement Learning in Pixel-based Environment” *International Conference on ICT Convergence (ICTC)*, 2024.

Projects

Development of Autonomous Manipulation and Grasping Based on Visual-Tactile sensing Imitation Learning Jan 2024 - Dec 2024, Jul 2025 - Present

- Developed a MuJoCo simulation environment and robot control interfaces for manipulation experiments.
- Trained a reinforcement learning policy for object relocation using simulated robot interactions.
- Applied imitation learning to teach a robot to position and orient objects for barcode scanning.
- Collected expert demonstrations via Apple Vision Pro teleoperation.

Development of Evaluation Methods to Support Skilled Robotic Manipulation and Grasping Performance Jul 2025 - Present

- Evaluated imitation learning algorithms on a peg-in-hole insertion task to compare their manipulation performance.
- Developed a tactile-guided residual reinforcement learning policy to refine a vision-based imitation policy during connector insertion.

Skills

Programming Language: Python, C, MATLAB

Tools: JAX/Flax, Pytorch, Tensorflow, MuJoCo, IsaacGym, Isaac-Sim (IsaacLab), ROS1 (neotic), ROS2 (Jazzy), Git

Modeling: AutoCAD, SolidWorks

Robots: Franka Research 3, Doosan manipulator, xArm, Puloon Symphony5, Allegro hand, Robotiq gripper, Inspire hand, Husky, Unitree Go2, Unitree G1

Sensors: XELA tactile sensor, AIDIN F/T sensor

Language: Korean(Native), English(Conversational)

Coursework

AI [Deep Reinforcement Learning \(CS285\)](#), Advanced Machine Learning Theory, Robot Learning

Math & Optimization	Engineering Mathematics, Optimization Theory, Random Processes
Robotics	Robotics, Probabilistic Robotics, Autonomous Driving Technology
Control	Mobile Control, Control Systems Engineering, Intelligent Control, Advanced Servo Control

Scholarships and Awards

Academic Excellence Scholarship Awards	Spring 2020, Spring 2021, Fall 2021
Outstanding Award at the Graduation Exhibition	Fall 2020

Teaching Experience

Teaching Assistant	Mar 2023 - Jun 2023
Optimization, Numerical Analysis (Instructors: Jun Moon)	